

Clinical Pharmacokinetics

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Objectives

Under the basic concepts of pharmacokinetics

Define important pharmacokinetic parameters

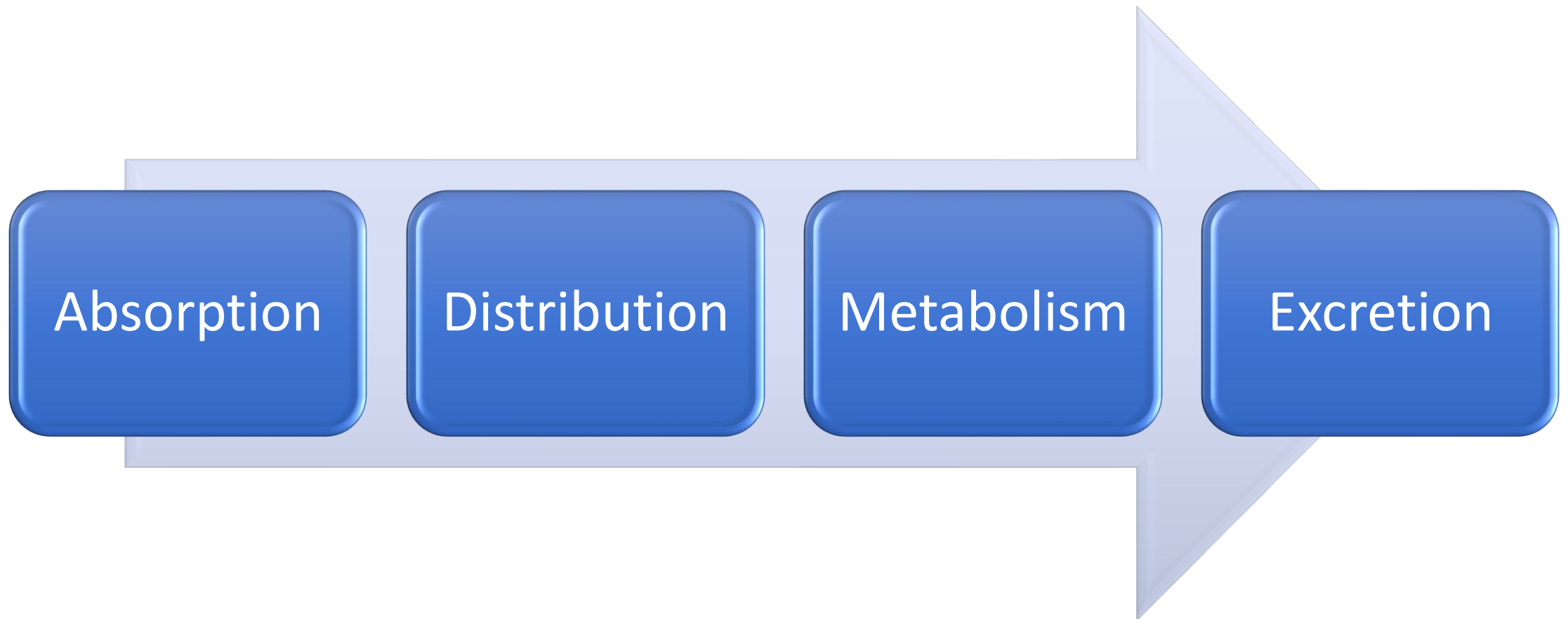
Introduce different compartmental models

Understand the design of therapeutic dosing and frequency

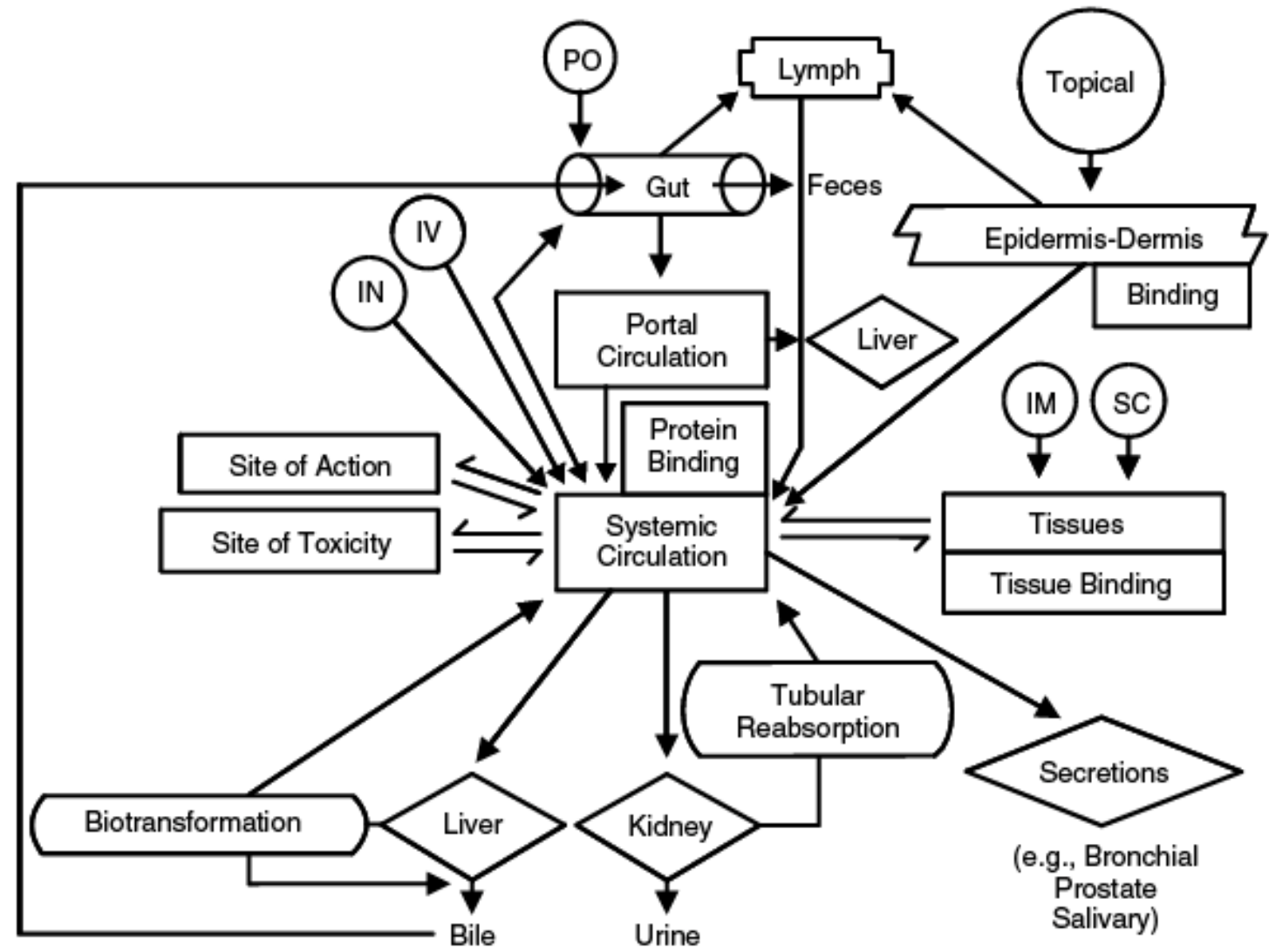
Understand the rationale of therapeutic drug monitoring

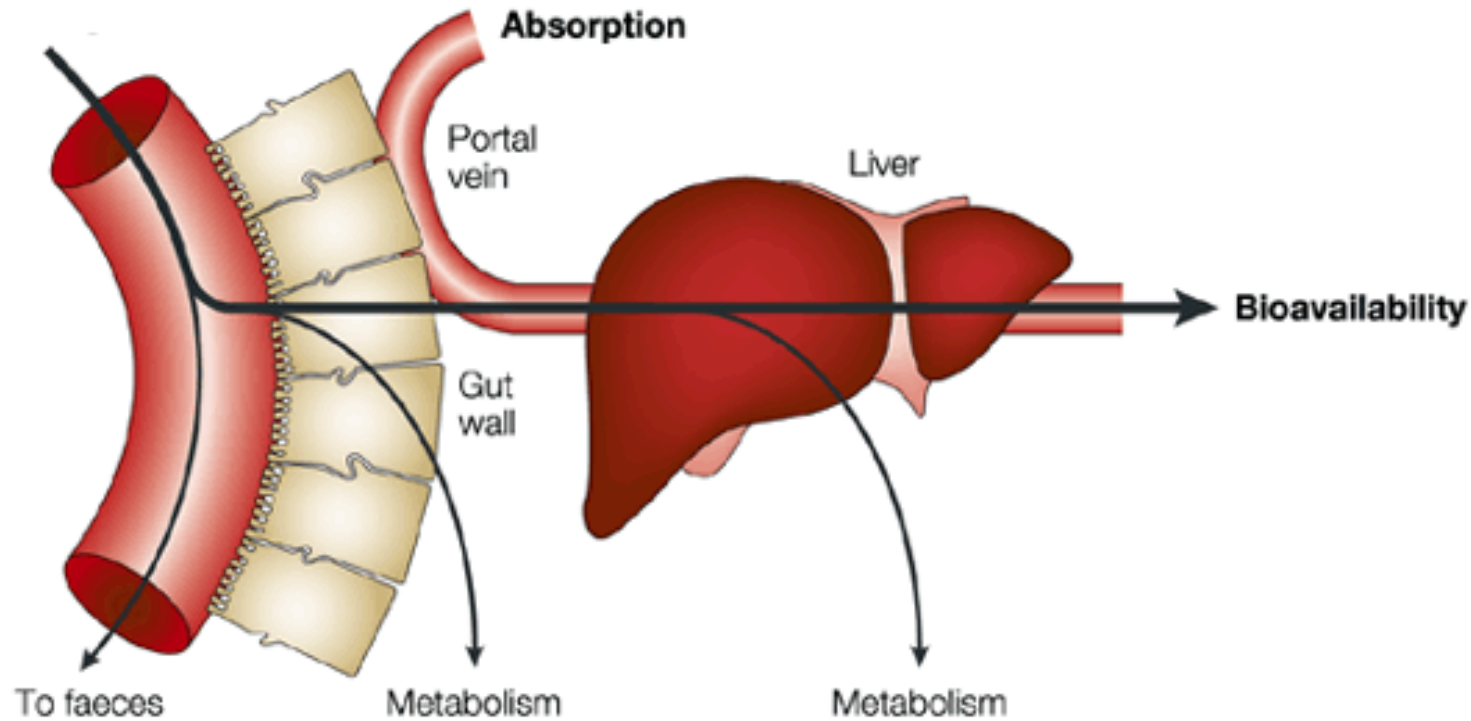
Basic concept of Pharmacokinetics

- Pharmacokinetics describes what our body does on the drug



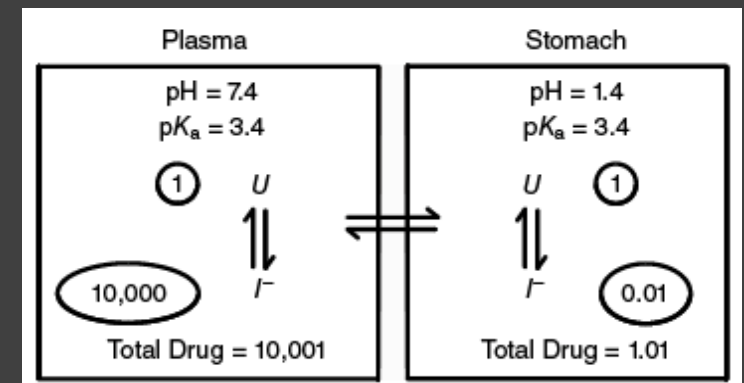
An overview of drug deposition

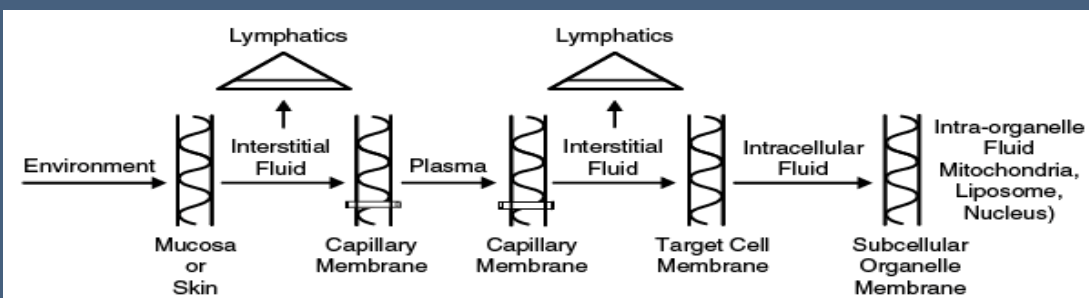
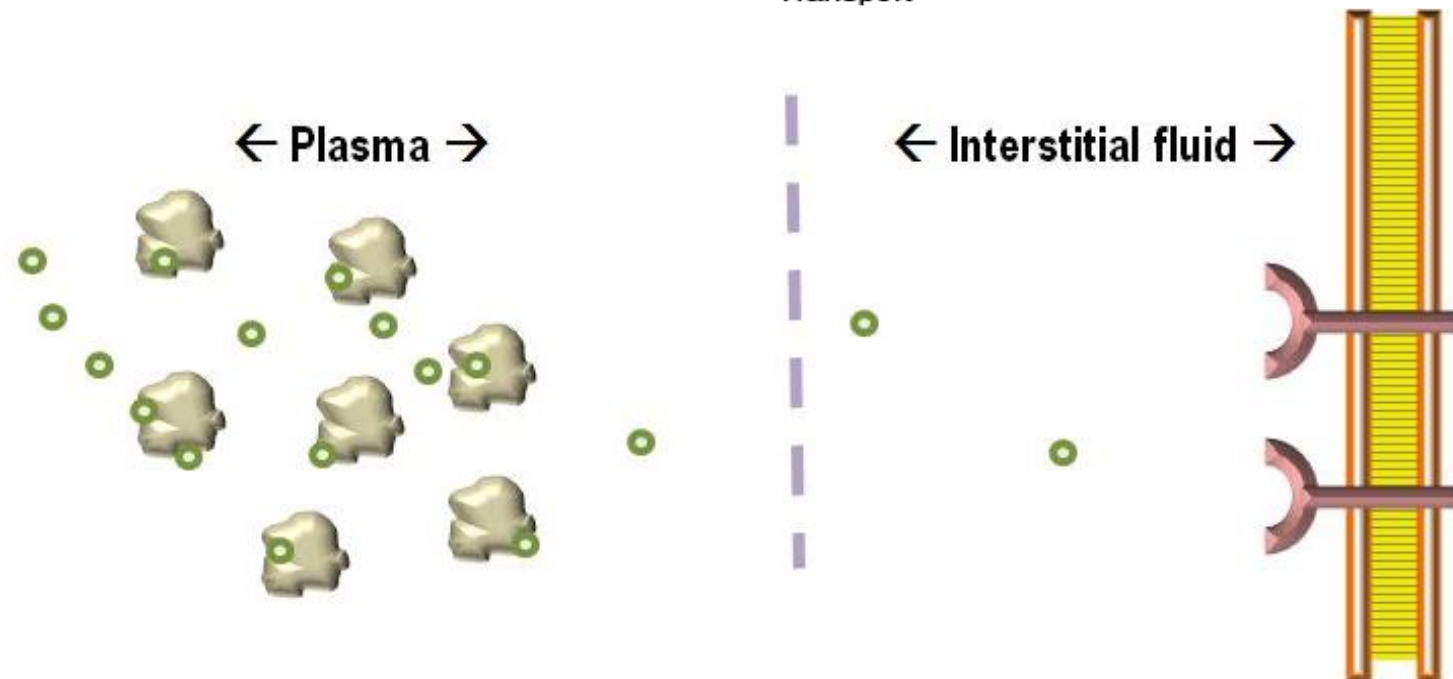
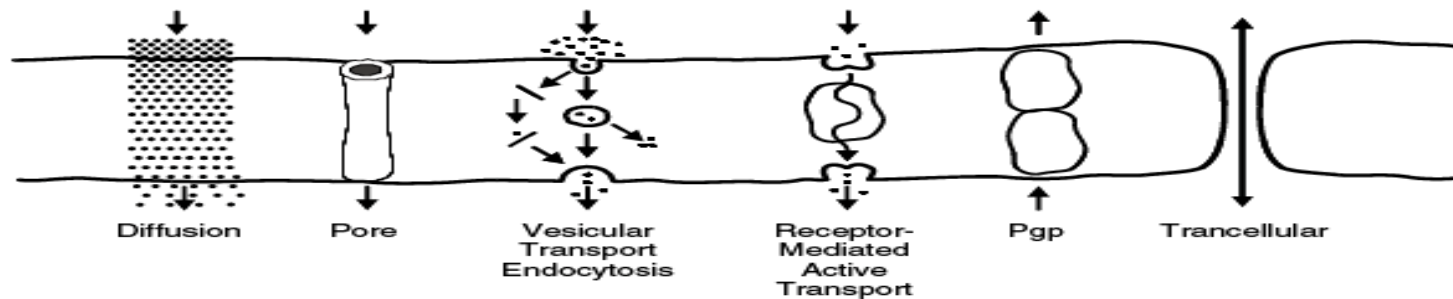




Basic concept of Pharmacokinetics

- Absorption
 - Route of administration
 - Absorption environment
 - Dosage form (modified release, extended release)
 - Physiological property



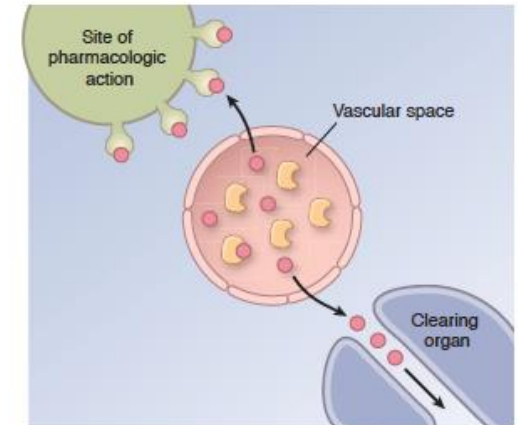
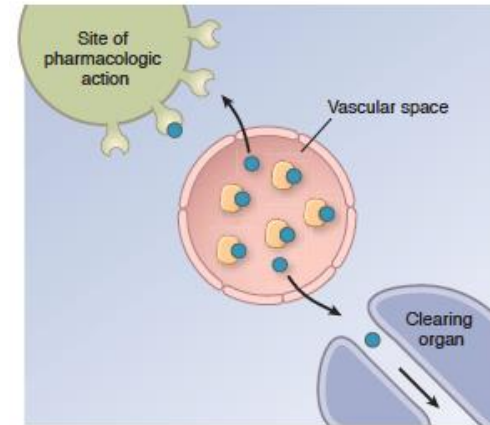


Basic concept of Pharmacokinetics

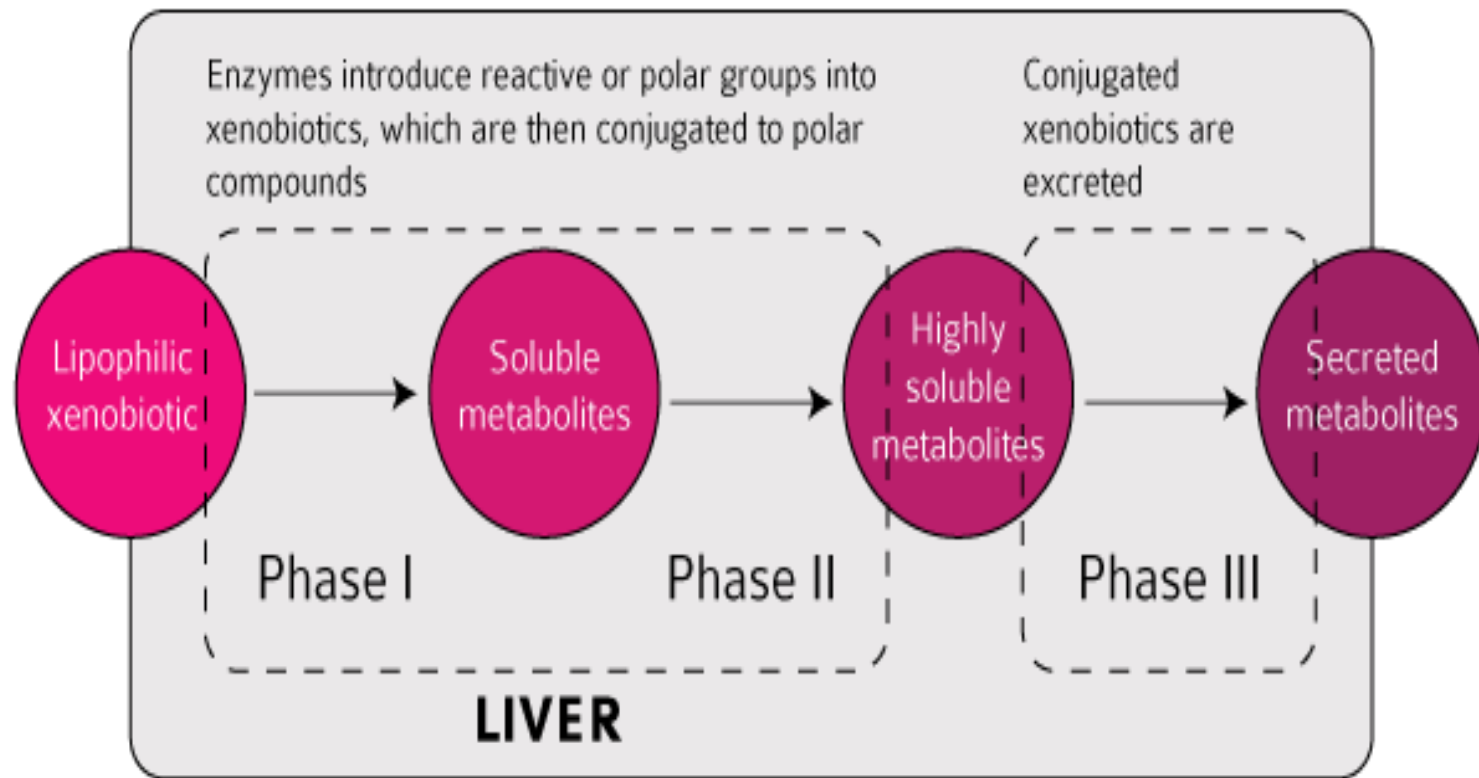
- Distribution
 - Binding to plasma protein
 - Physiological volume
 - Membrane permeability
 - Preferential tissue perfusion

Drug distribution

Organ	Blood flow (ml/min)	Organ mass (kg)	Normalized blood flow (ml/min/kg)
Liver	1700	2.5	680
Kidneys	1000	0.3	3333
Brain	800	1.3	615
Heart	250	0.3	833
Adipose	250	10	25
Others	1400	55.6	25
Total	5400	70	



Compartment	Examples
Total body water	Small water soluble molecules (ethanol)
Extracellular water	Large water soluble molecules (mannitol)
Blood plasma	Highly plasma protein bound molecules Very large molecules Highly charged molecules (heparin)
Adipose tissue	Highly lipid soluble molecules (diazepam)
Bone and teeth	Certain ions (fluoride, strontium)



Basic concept of Pharmacokinetics

- Metabolism
 - Activate pro-drug or de-activate active drug
 - Mostly occurs in the liver by Phase I/II reactions
 - Dependent on circulation, organ function, genetic variability and drug-drug interaction

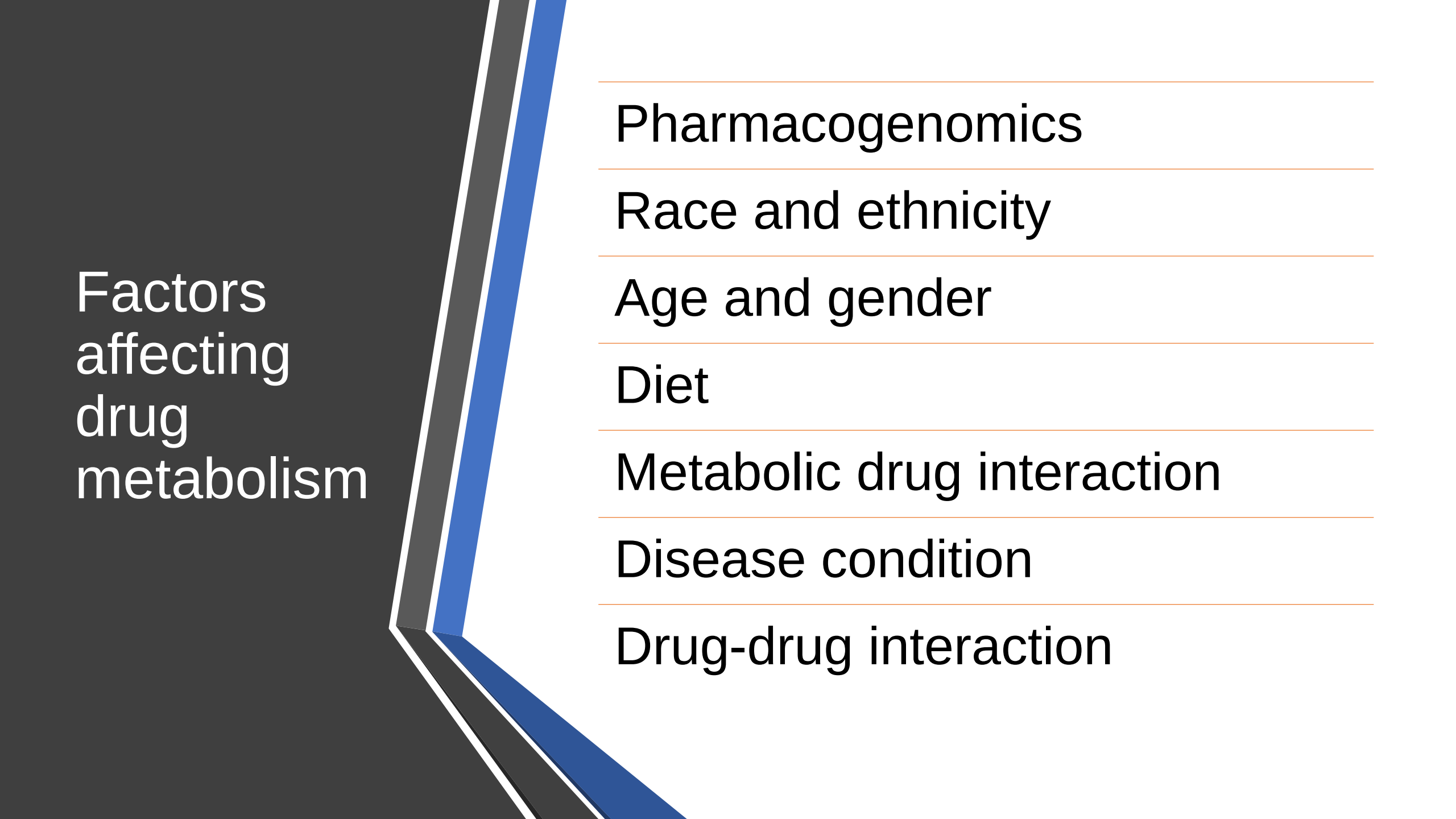
Drug metabolism

Phase 1 metabolism (Cytochrome P450)

- Oxidation
- Reduction
- Hydrolysis

Phase 2 metabolism (Transferase reaction)

- Glucuronidation
- Acetylation
- Methylation
- Glycine conjugates
- Glutathione conjugates
- Sulphate conjugates



Factors affecting drug metabolism

Pharmacogenomics

Race and ethnicity

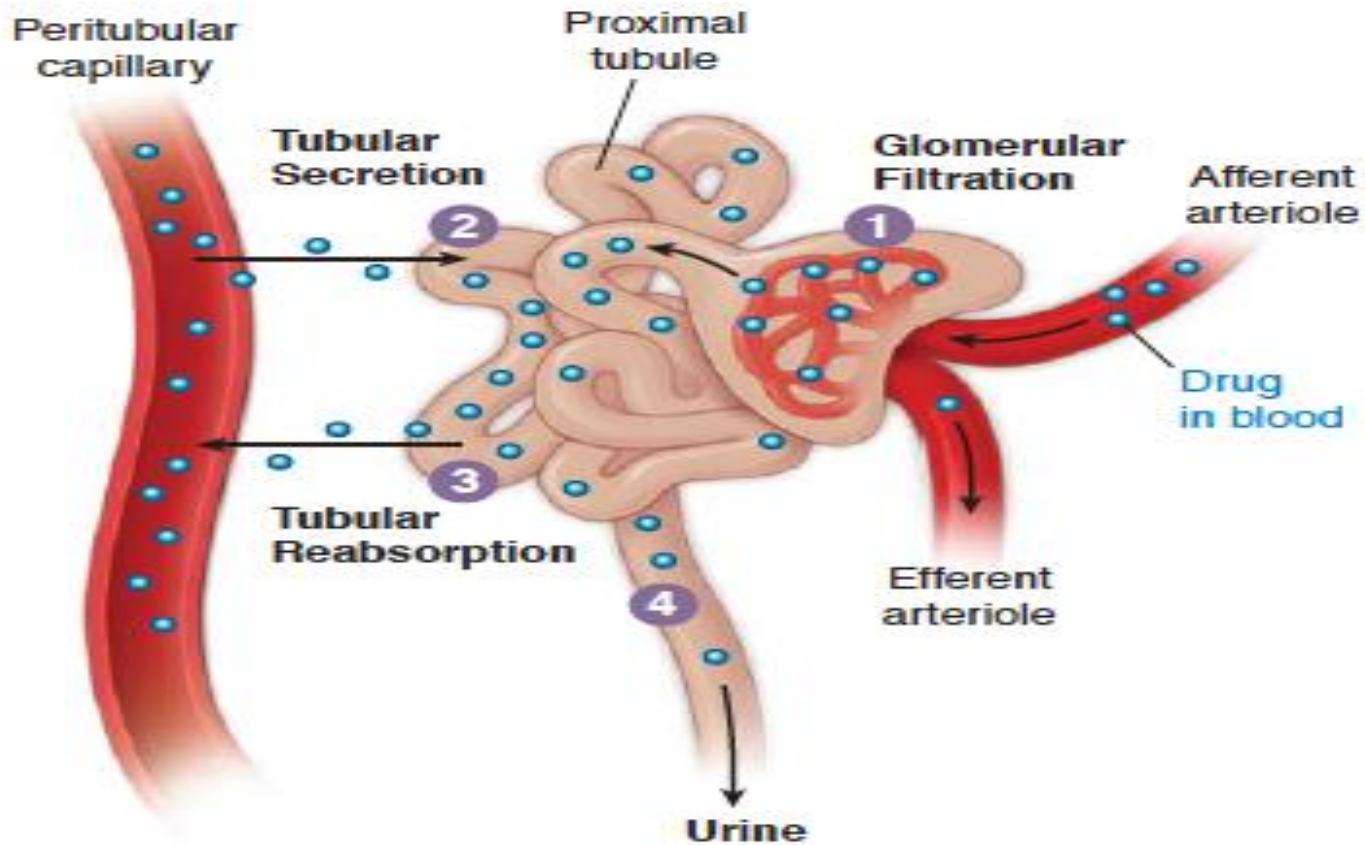
Age and gender

Diet

Metabolic drug interaction

Disease condition

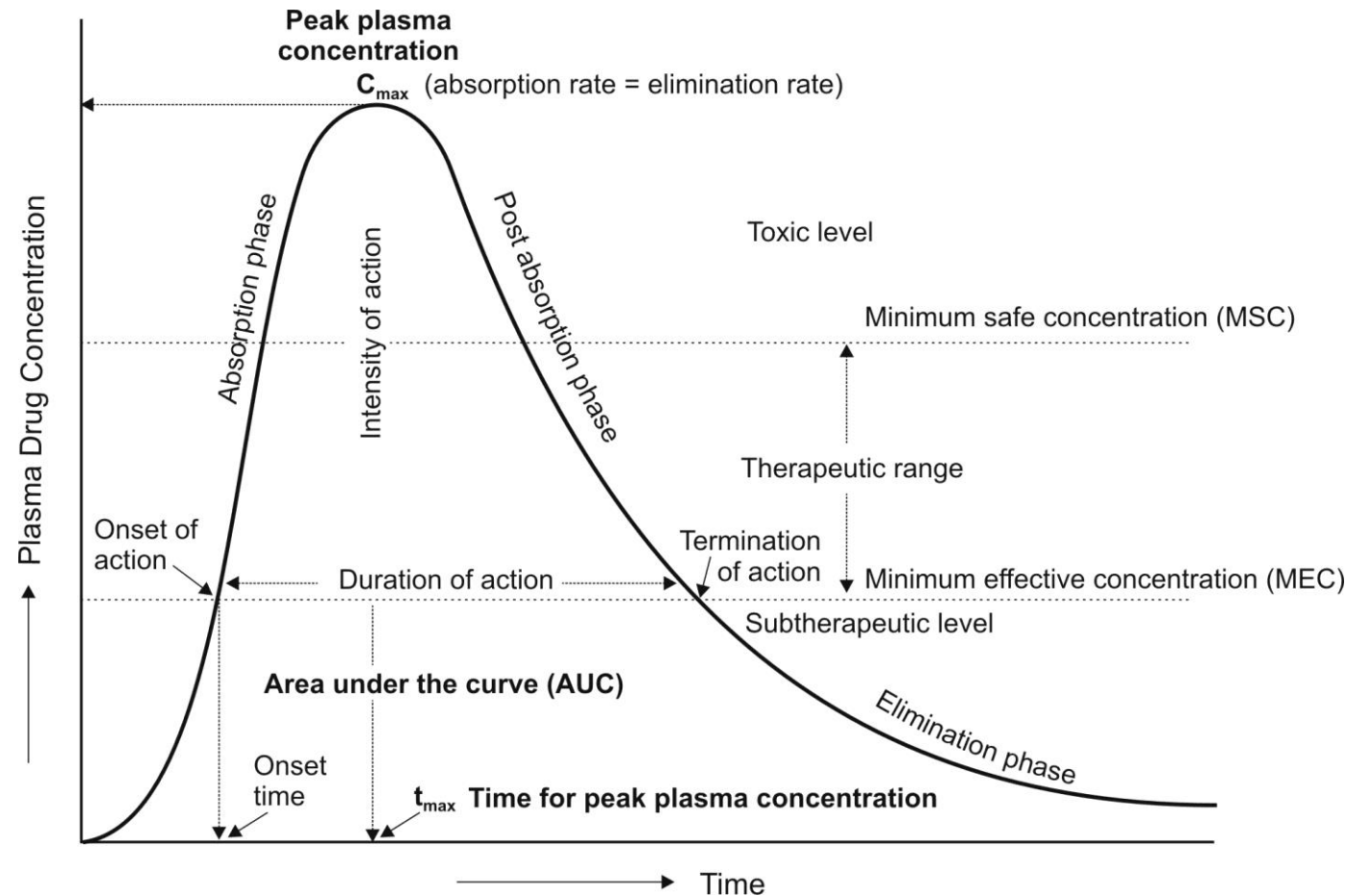
Drug-drug interaction



- Excretion
 - Urinary
 - Biliary
 - Fecal
 - Others e.g. breast milk

Basic concept of Pharmacokinetics

Concentration time curve

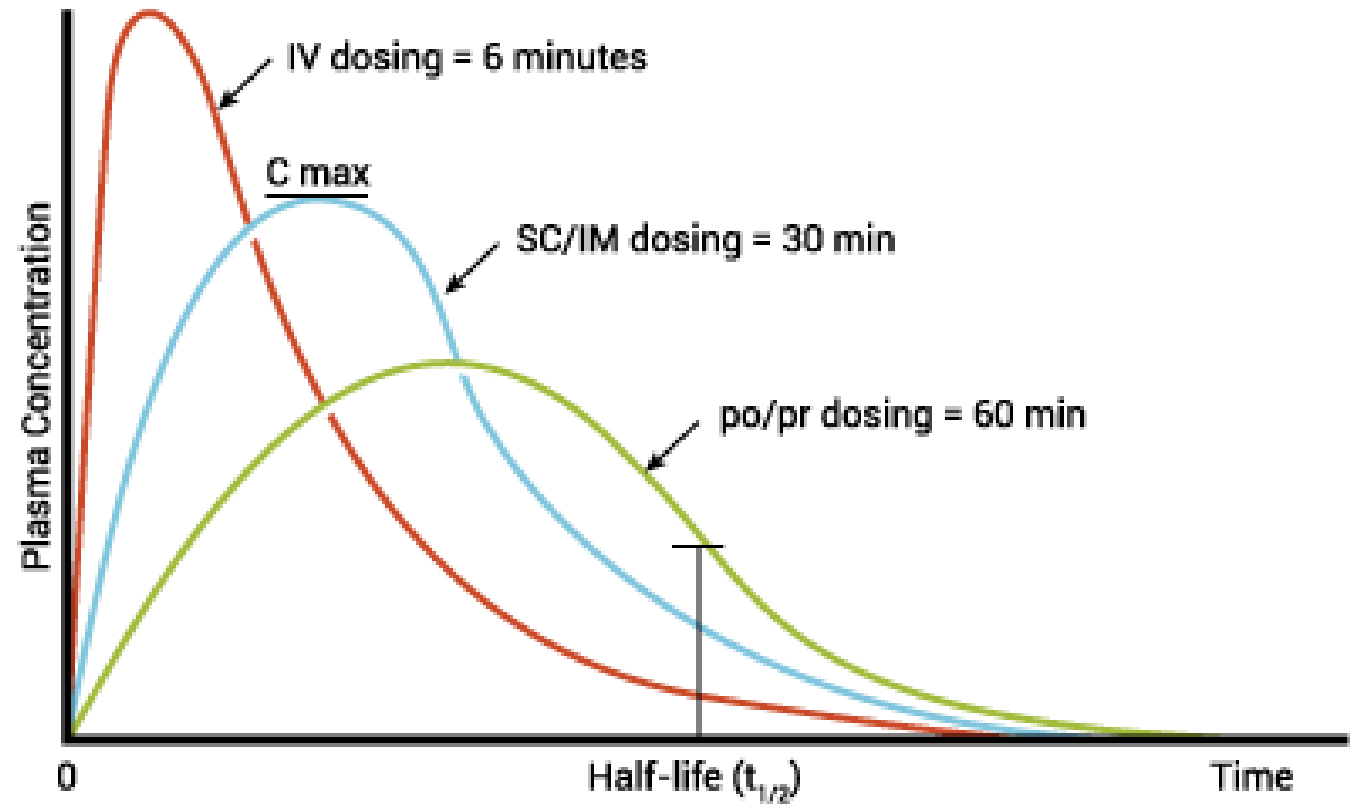


Compartmental Model

- First order kinetics
- Zero order kinetics
 - Only when saturation occurs would non-linear behavior become evident

Number of T1/2	% of drug eliminated	% of drug remaining
1	50	50
2	75	25
3	87.5	12.5
4	93.75	6.25
5	96.88	3.12
6	98.44	1.56
7	99.22	0.78
8	99.61	0.39
9	99.80	0.19
10	99.90	0.10

Concentration time curve



Pharmaco-
kinetics
parameters

C_{max} (Maximum concentration)

t_{max} (Time to achieve C_{max})

AUC (Area under curve)

F (Bioavailability)

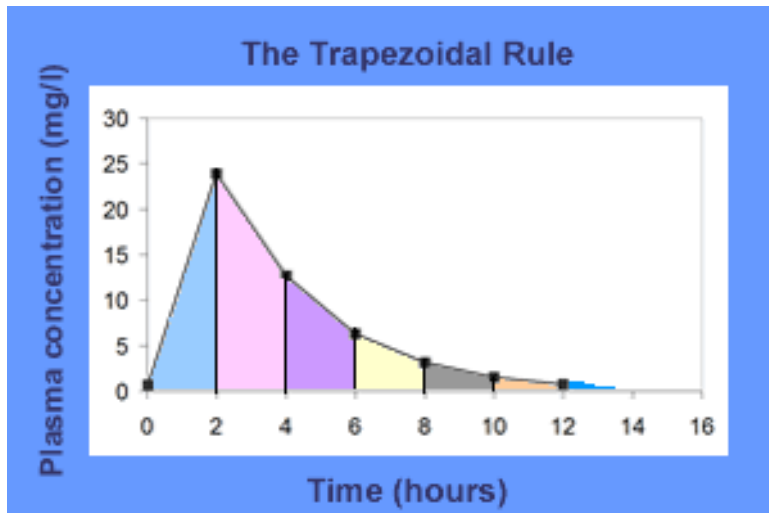
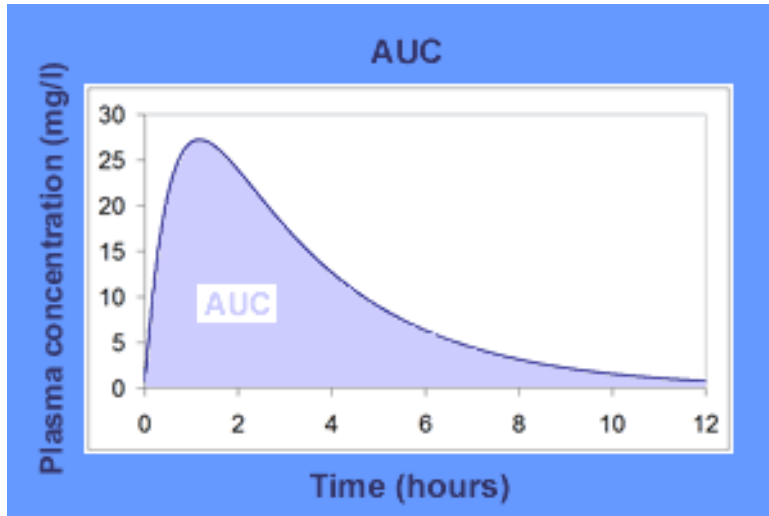
V_d (Volume of distribution)

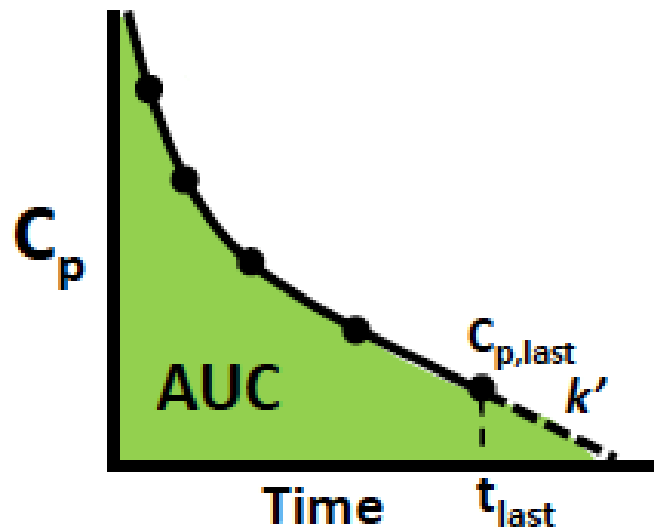
CL (Clearance)

$t_{1/2}$ (elimination half life)

Area under curve

- Reflects the actual body exposure to drug after administration
- Dependent on the rate of elimination of the drug and the dose administered
- Directly proportional to the dose when the drug follows linear kinetics

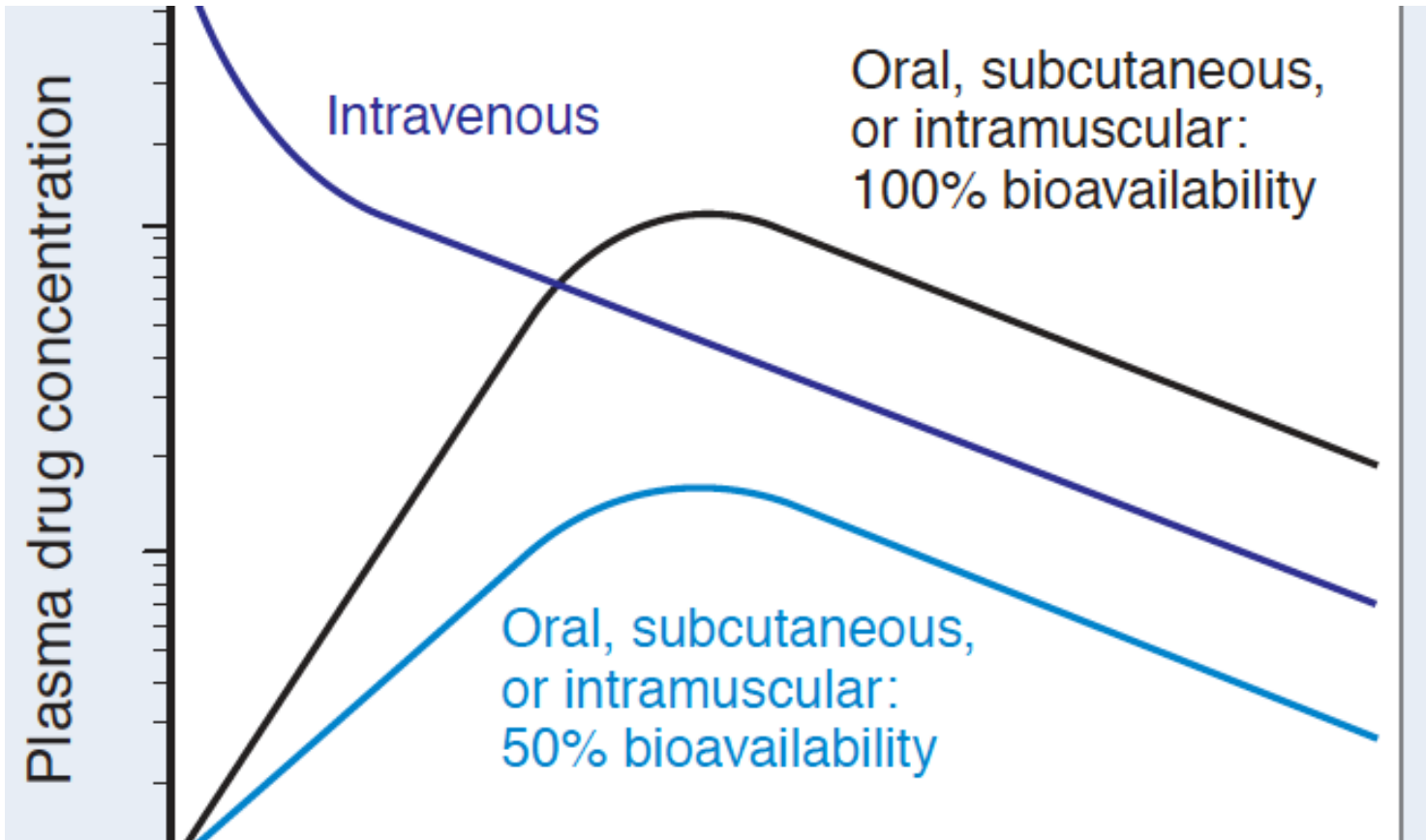




$$AUC = \underbrace{AUC_{0-t_{last}}}_{\text{Calculated Interval Areas (Trapezoidal Rule)}} + \underbrace{\frac{C_{p,last}}{k'}}_{\text{AUC}_{(last-\infty)} \text{ (Extrapolate from } t_{last} \text{ to } t=\infty)}$$

where k' is the terminal slope from semi-log graph of C_p vs. time

Area under curve

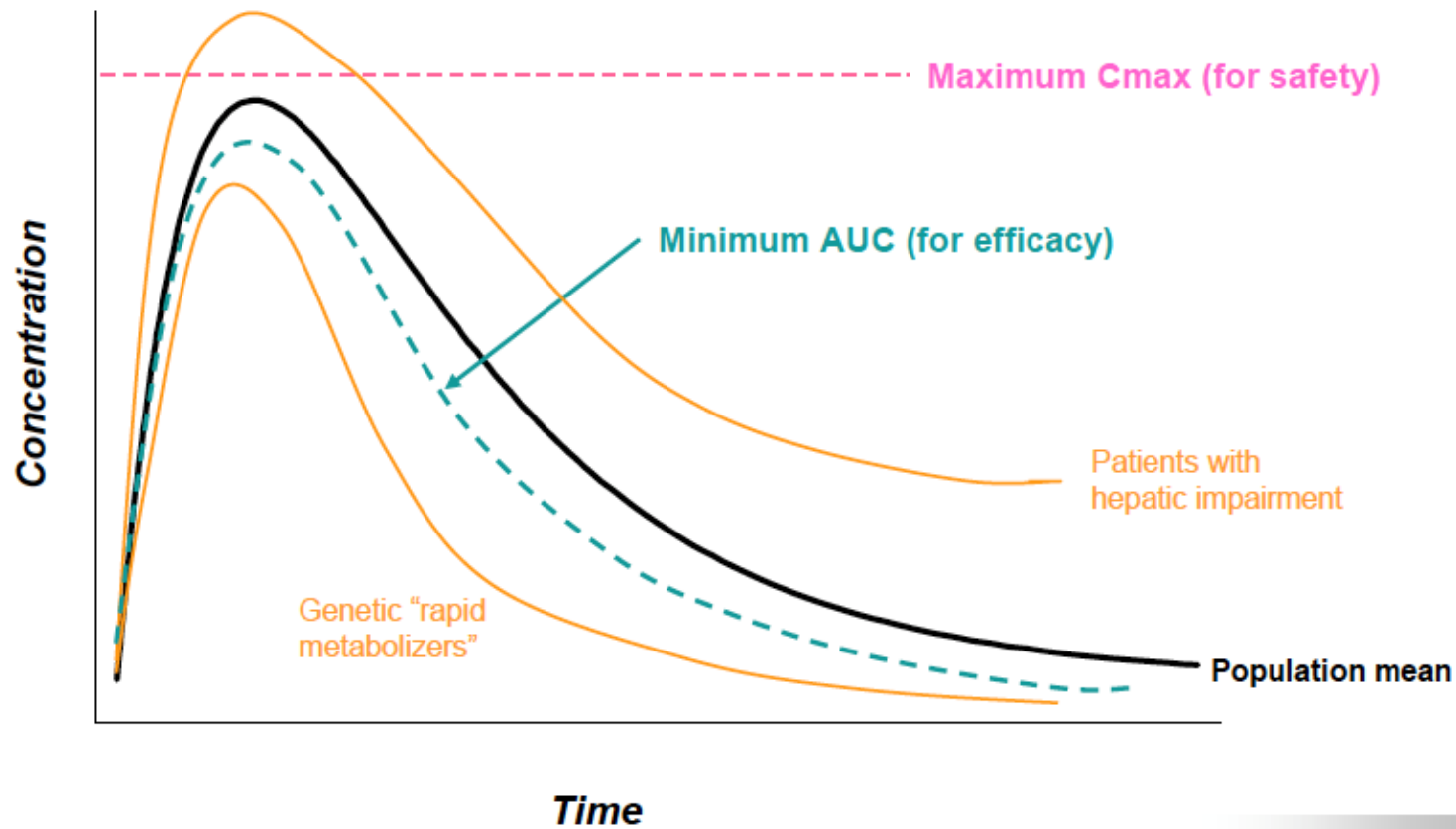


Utility of AUC

Bioavailability (F)

- Fraction of the dose available to the systemic circulation
- Factors influencing bioavailability
 - Absorption environment (health of the GI tract, pH, food effect)
 - First pass effect (genetic, drug-drug interaction)

$$F = \frac{[AUC_{po} \times Dose_{iv}]}{[AUC_{iv} \times Dose_{po}]}$$



Clearance

- Limits the time course of action of the drug at its molecular, cellular and organ targets
- $CL = [\text{Metabolism} + \text{Excretion}] / \text{Plasma drug conc.}$
- Factors influencing clearance
 - Metabolism
 - Excretion

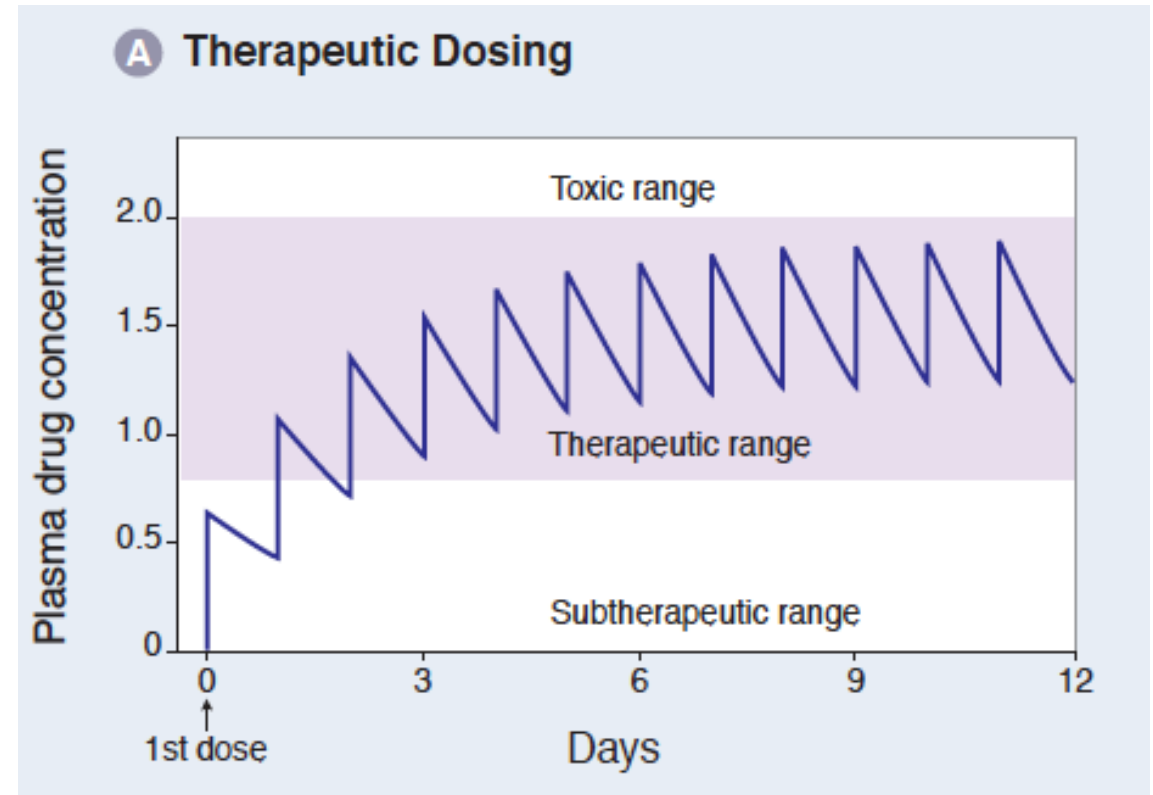
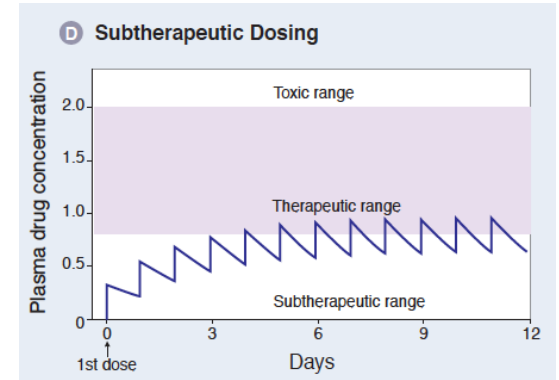
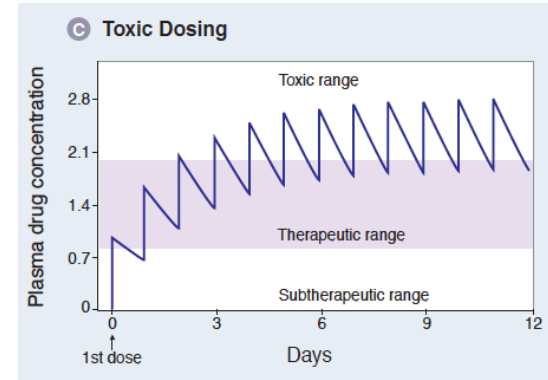
$$CL = \text{Dose}_{iv} / AUC \text{ OR}$$

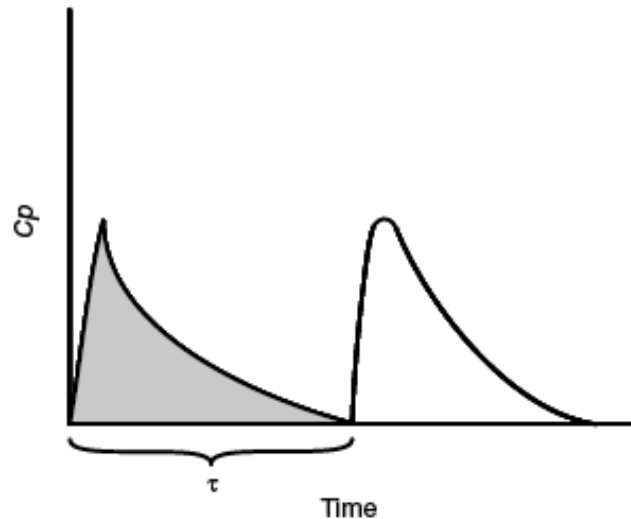
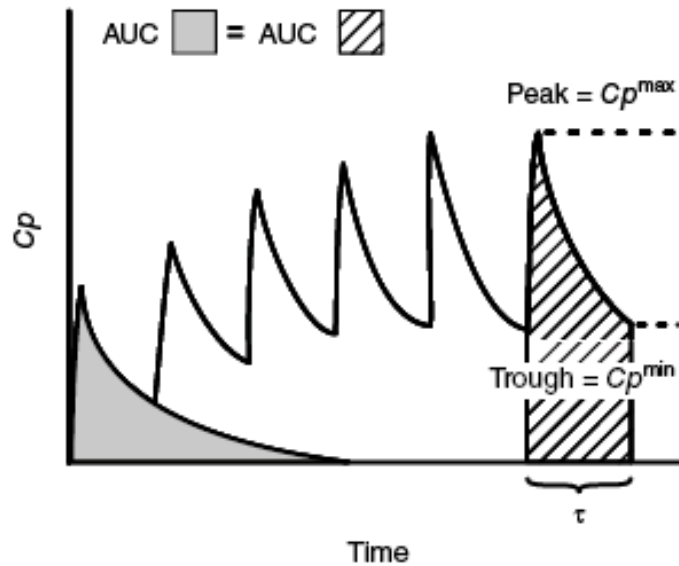
$$CL = Fx\text{Dose}_{po} / AUC$$

Utility of AUC

Therapeutic dosing and frequency

- Dose
 - Potency
 - Bioavailability
 - Volume of distribution
- Frequency
 - Elimination rate
- Maintain the peak plasma concentration below the toxic concentration, and the trough drug level concentration above the minimally effective level



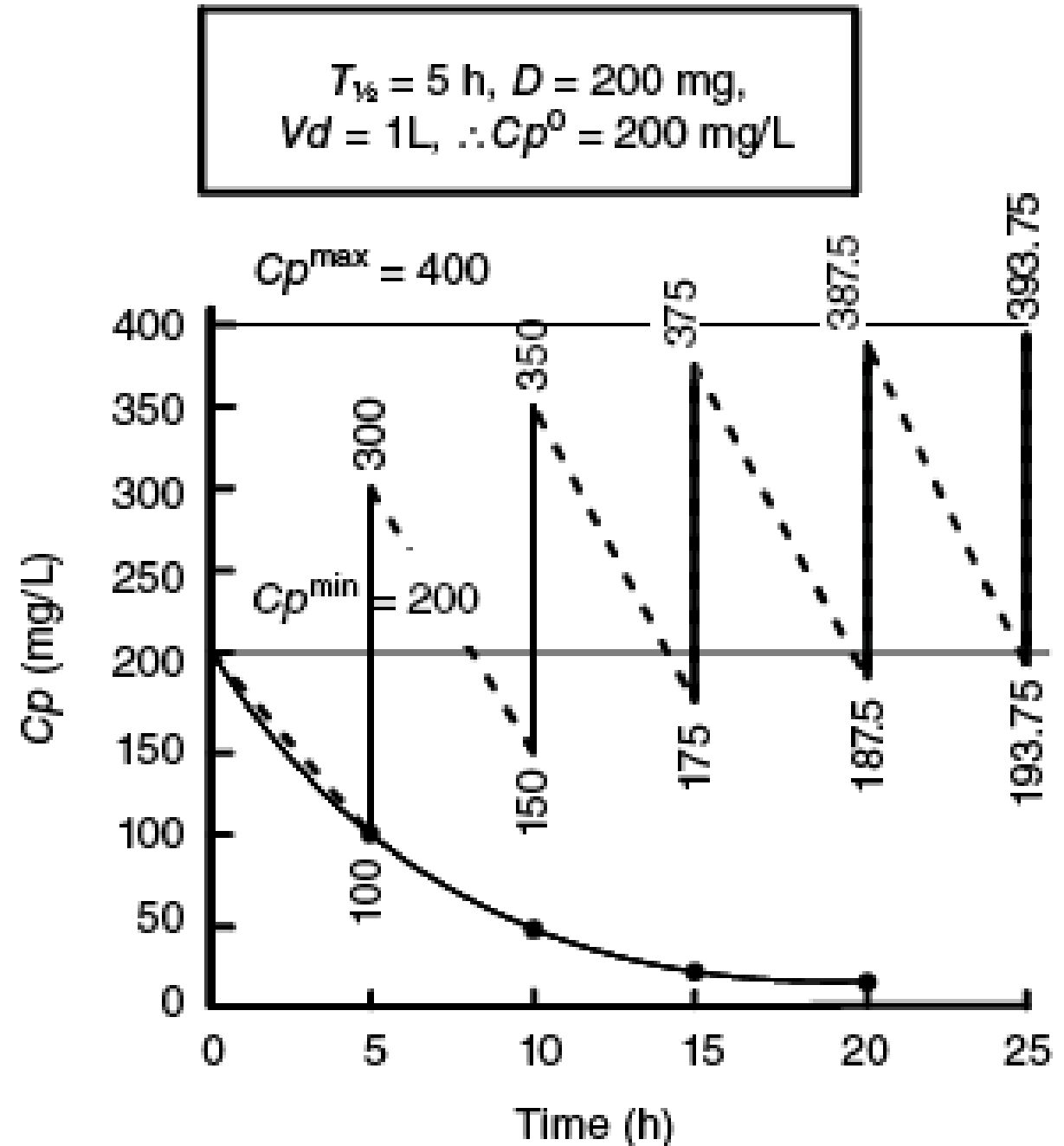


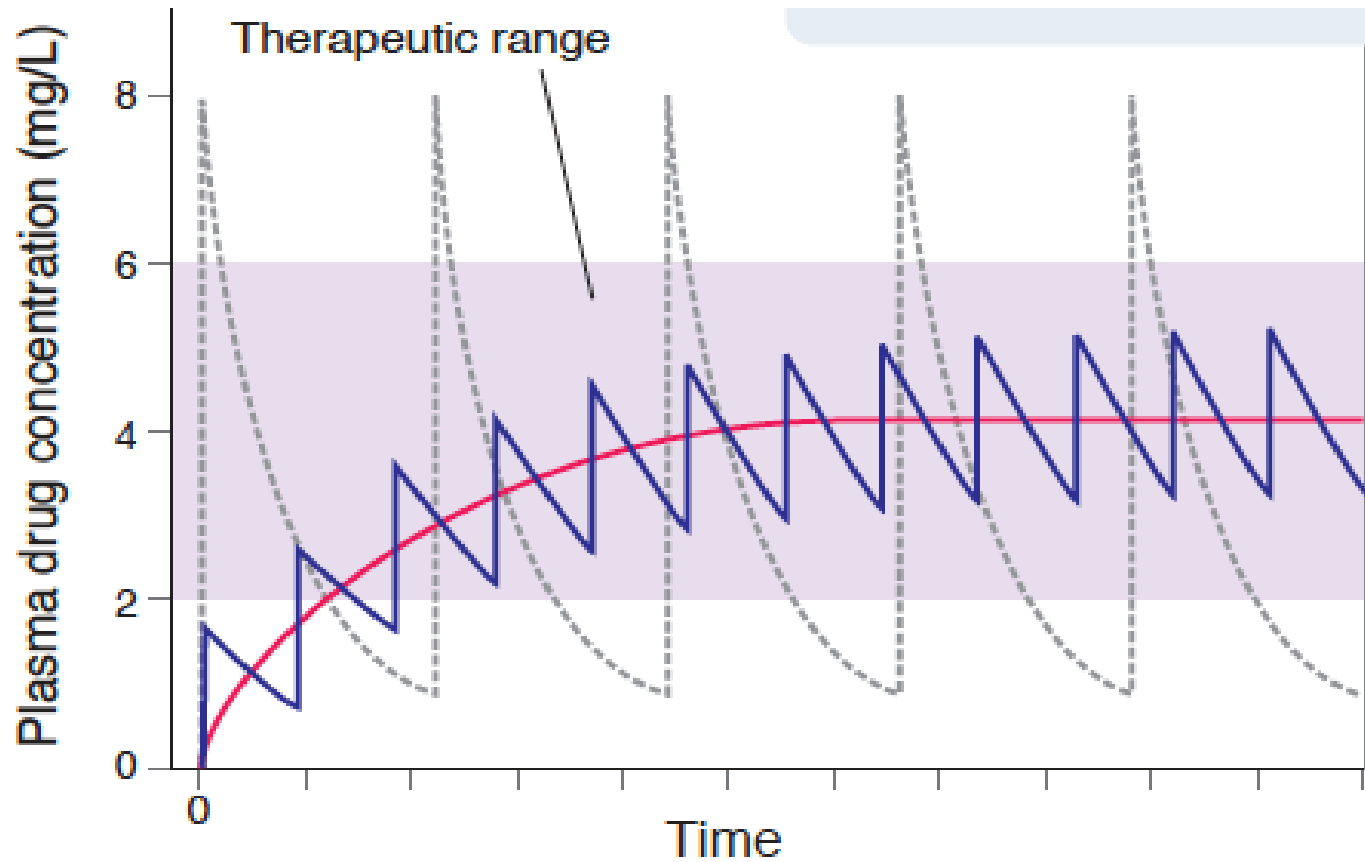
Therapeutic dosing and frequency

- If a second dose of drug were given after the first dose was completely eliminated (approx. 5 $T_{1/2}$) then this dosage regimen does not result in any drug accumulation in the body

Therapeutic dosing and frequency

- Steady state is defined as the dosing interval in which the AUC for that interval is equal to the single-dose AUC.
- Loading dose = $C_{p\max} \times V_d$





Therapeutic dosing and frequency

- Optimal dosing regimens typically maintain the steady state plasma drug concentration within the therapeutic window
- As a rule of thumb, steady state can be achieved after 5 elimination half lives
- $C_{\text{steady state}} = \frac{\text{Bioavailability} \times \text{dose}}{\text{interval} \times \text{clearance}}$

Rationale of giving a loading dose

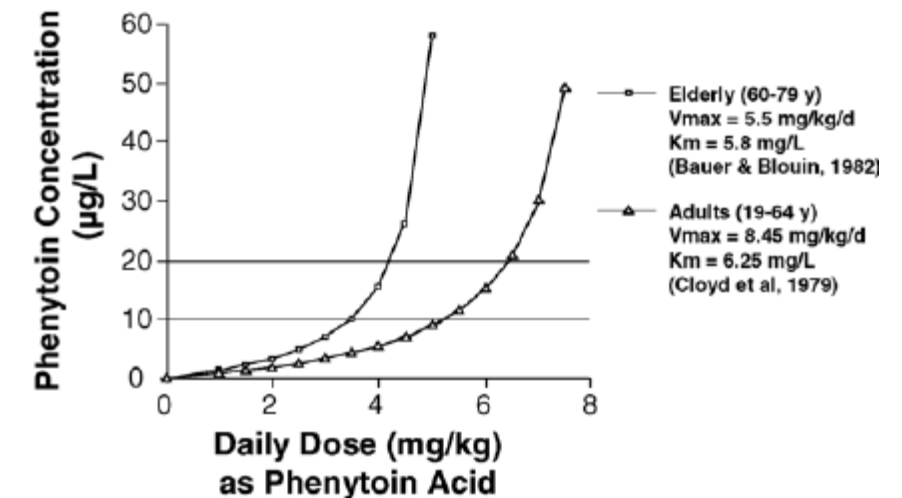
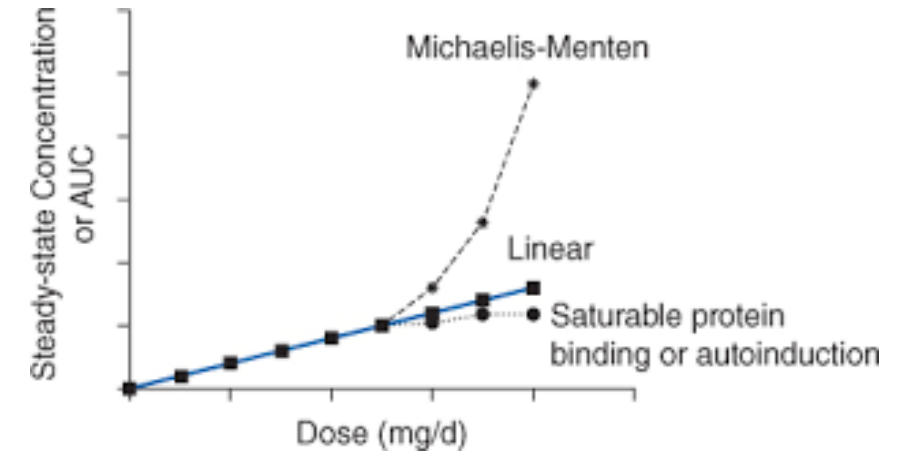
A drug does not show therapeutic activity unless it reaches the desired steady state

It takes about 4-5 half lives to attain the steady state and therefore time taken will be too long if the drug has a long half-life.

Plateau can be reached immediately by administering a dose that gives the desired steady state instantaneously before the commencement of maintenance dose

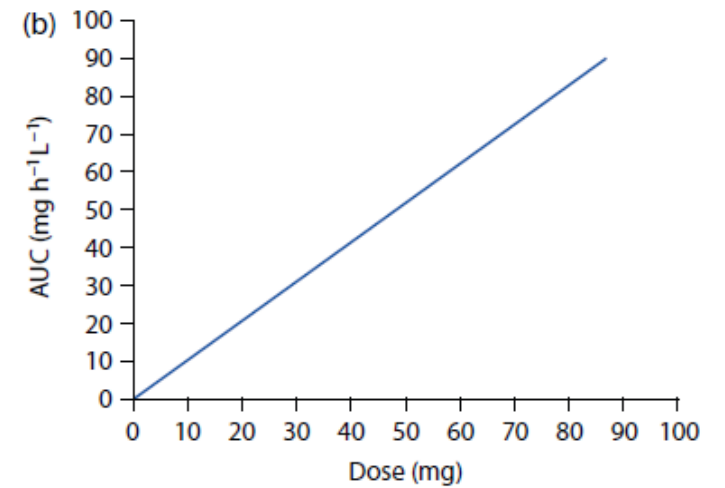
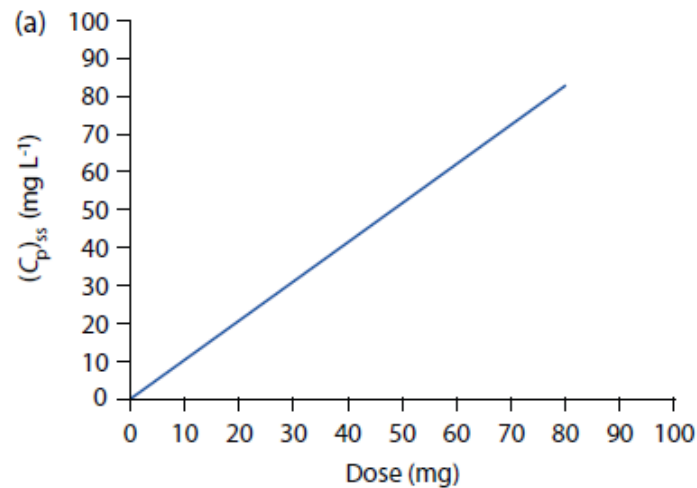
Phenytoin PK parameters

- Bioavailability: oral 70-100%
- Volume of distribution: 70L
- Clearance: 10L/h
- Elimination half-life: avg 22hrs (7-42hrs)
- $T_{1/2} = (0.693 \times V_d) / CL$
- $T_{1/2} = (0.693 \times 70) / 10 = 6.74\text{hrs}$



Linear pharmacokinetics

- Pharmacokinetic parameters are not expected to change when different doses are administered and/or when the drug is administered via different routes as a single dose or multiple doses
- The kinetics is characterized by the FIRST ORDER KINETICS
- Plasma concentration at a given time at steady state and the area under curve will both be proportional to the dose administered



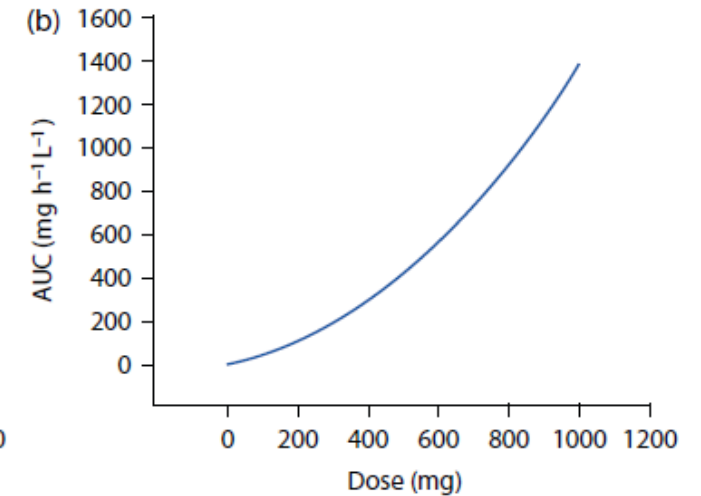
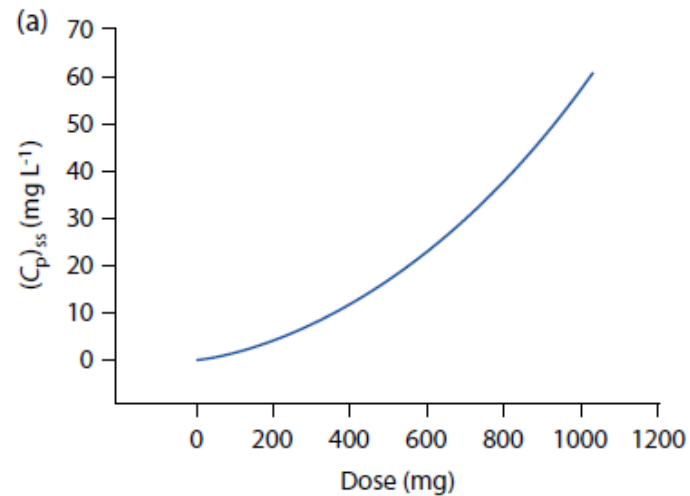
Nonlinear pharmacokinetics

- First order kinetics greatly simplifies dosage design, bioavailability, dose-response relationship, prediction of drug distribution and disposition
- First-order elimination applies only to compounds that are eliminated exclusively by mechanisms not involving enzymatic or active transport processes
- At clinical dosages, the majority of drugs do not reach saturation concentrations at the reaction sites



Nonlinear pharmacokinetics

- The pharmacokinetic parameters may vary depending on the administered dose
- Other kinetic may be involved e.g. ZERO ORDER KINETICS
- Plasma concentration at a given time at steady state and the area under curve will NOT be proportional to the dose administered



Processes leading to nonlinear pharmacokinetics

Absorption

- Extent of absorption of amoxicillin decreases with an increase in dose

Distribution

- Plasma protein binding of disopyramide is saturable at the therapeutic concentration, resulting in an increase in the volume of distribution with an increase in dose of the drug

Metabolism

- Both phenytoin and ethanol have saturable metabolism, which means that an increase in dose results in a decrease in hepatic clearance and a more than proportional increase in AUC

Excretion

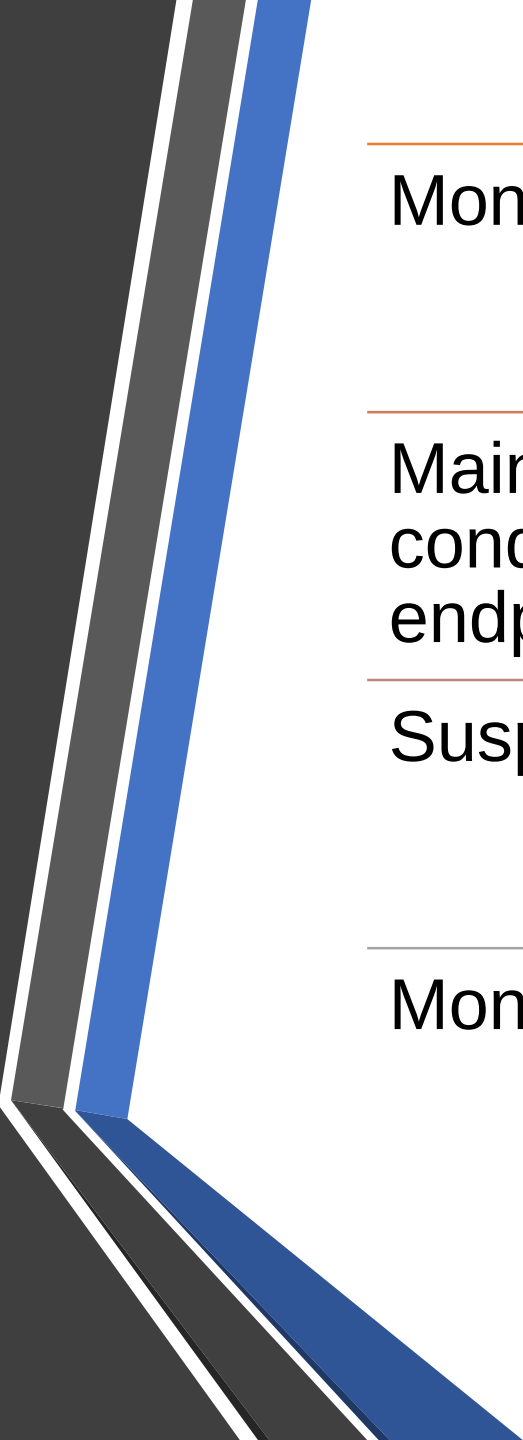
- dicloxacillin has saturable active secretion in the kidneys, resulting in a decrease in renal clearance as dose is increased

Therapeutic drug monitoring

Clinical practice of measuring a specific drug at designated intervals to maintain a constant concentration

Involves not only measuring drug concentrations, but also the clinical interpretation of the result

Aims at optimizing an individual's dosage regimen



Indications for requesting plasma drug concentration

Monitor drug compliance

Maintain drug efficacy especially for
conditions with poorly defined clinical
endpoints

Suspected toxicity

Monitor and detect drug interaction

Which drugs require therapeutic drug monitoring?

Narrow therapeutic range

Established target concentration range

Significant intra-/ inter-subject variability

Significant dose response relationship

Nonlinear pharmacokinetic profile

Availability of cost effective drug assay

Which drugs
do not
require
therapeutic
drug
monitoring

Clinical outcome is unrelated either
to dose or plasma concentration

Dosage need not be individualized

The pharmacological effects can be
clinically quantified

Wide therapeutic index

Drugs which require therapeutic drug monitoring

Digoxin

Lithium

Phenytoin

Cyclosporin

Tacrolimus

Sirolimus